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Multimodal scene-graph matching for cheapfakes detection

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Abstract

The development of technology and social media platforms has led to the proliferation of fake news, including the cheapfakes problem. Cheapfakes can be produced easily and spread quickly; a common type is out-of-context misinformation. It is worth investigating a proper algorithm that can efficiently detect such types of misinformation. In this work, we study a new approach to the detection problem of cheapfakes by using graphical neural networks to detect out-of-context samples using the dataset from the ICME 23 Grand Challenge on Detecting Cheapfakes. Specifically, our model utilizes scene graph matching and a language model pre-trained for the natural language inference task to solve task 1 of the challenge. We also propose effective methods to generate labeled data from an unlabeled training set of the challenge. Our proposed method achieved an F1-score of 85.69% and an accuracy score of 85.50% on the public testing set, surpassing the baseline by 4.69% and 3.60%, respectively.

Keywords: Cheapfakes, Scene graphs, Graph embedding, Misinformation

1 Introduction

Nowadays, the development of technology, especially social networking platforms like Facebook and Twitter, has helped people connect with each other and update information quickly. However, it also leads to the proliferation of fake news, including deepfakes and cheapfakes. Cheapfakes are created using simple tools, with a typical example being out-of-context (OOC) misinformation, while the creation of deep fakes is more advanced with the help of deep learning. Although deepfakes are more realistic and convincing, cheapfakes are more prevalent as they are simpler to produce and can spread quickly on the Internet. For this reason, one of the well-known competitions during the last few years, the ICME 2023 grand challenge on detecting cheapfakes[1], focuses on identifying OOC samples. The challenge consists of two tasks: Task 1 requires finding miscontextualization in the image-caption triplets, while Task 2 requires determining if a given image-caption pair is authentic or fake.

For this challenge, many methods have relied on the COSMOS baseline [1]. However, these methods and the COSMOS baseline utilize the detected visual objects and encoded sentences

[1]https://detecting-cheapfakes.github.io/

to detect cheapfakes, but they have not considered the relationships between objects. Moreover, many of these methods use part of the test data to finetune the models, so they are prone to overfitting since the test size is small. Therefore, we propose a new method that utilizes scene graph matching and natural language inference (NLI) techniques to achieve comparable results to the COSMOS baseline on task 1 without finetuning on the testing set to avoid overfitting. Scene graphs can represent objects and their complex relationships in visual and textual data, providing more insights into the data. To the best of our knowledge, this is the first time they have been used to detect cheapfakes.

This paper aims to use scene graphs and natural language inference (NLI) techniques to solve the cheapfakes detection challenge task 1. Scene graphs have been widely applied for image retrieval tasks [2–5] and have achieved excellent results. In this work, we use various methods to extract visual and textual scene graphs and utilize the scene graphs to calculate the similarity between images and captions. We also integrate an NLI module into the framework to learn the semantic relations between the pairs of captions. The train set covers diverse topics and is much larger than the testing set but is unlabeled, so we propose new techniques to generate labeled training data. Experimental results on the challenge's public test dataset demonstrated the effectiveness of the proposed method.

The paper can be organized as follows. In Section 1, we introduce the main problem and briefly summarize the contribution of our work. Section 2 presents the relevant works during recent years. In Section 3, we propose our main approach for the detection of cheapfakes. Section 4 demonstrates the experiments and compares the performance of our techniques with those of other works. The paper ends with a conclusion and further work.

2 Related work

There have been different works related to the problem of detecting cheap fakes during the recent years.

2.1 Cheapfakes detection

Detecting out-of-context misinformation has been extensively researched [1, 6–8], and sustained efforts have been made to develop solutions for it [9–17]. Many datasets have been developed for this problem, including the MAIM [18], Twitter-COMMs [19], NewsCLIPpings [20] and the COSMOS dataset. Among these, the COSMOS dataset and a self-supervised learning algorithm with the same name [1] was introduced on the Grand Challenge on detecting cheapfakes and have promoted research into this problem. Many later methods have relied on the COSMOS baseline, with several improvements proposed to enhance the results, including integrating a pretrained model for the Natural Language Inference (NLI) task [10–12] and using the Large Language model [13]. However, these methods rely on the testing set to finetune the models and improve their performance. Recently, some new methods that do not rely on the COSMOS baseline have been proposed [14, 15], but their performances are still low compared to the COSMOS baseline. Therefore, this paper proposes a new method that utilizes the features extracted from the scene graphs and the NLI models to produce results comparable to the COSMOS baseline without finetuning on the test dataset.

2.2 Scene graph generation

Typically, a scene graph can be determined as a structured representation of objects, their properties, and how they relate to each other [21]. Scene graph parsing can be divided into three major directions: visual scene graph [22–31], semantic scene graph [32–36], or extracting scene graphs from both modalities. Visual scene graph parsers utilize an object detection model, mostly Faster R-CNN [37], and a relationship recognition module to recognize the relationships between the objects. Parsing textual scene graphs applies various models like a rule-based model [32, 33], Bi-LSTM model [34] to Large Language model (LLM) [36], and these methods often depend on an intermediate representation like dependency graphs [34] or abstract meaning representation [35]. Scene graphs have been applied in various vision-language tasks that require high-level understanding about the images, including image

captioning [38–40], image retrieval [2, 3], and visual question answering [41, 42].

3 Methodology

This section presents how we can formulate the main problem and the proposed method for detecting cheapfakes.

3.1 Problem formulation

The problem takes a triplet containing an image and two captions as input. The aim is to predict whether the two captions associated with the image are out-of-context, so the output shall be a binary label. Out-of-context use of an image is defined as using the image as evidence for events that are either untrue or unrelated [1]. If two captions refer to the same object in the image but have different semantic meanings, like relating to different events, this suggests that the image is being used out of context. On the other hand, if the captions are associated with the same event, regardless of the objects they describe, it is considered as not out-of-context use of the image. The objective of this task is to identify the mis-contextualization, and the results do not determine the veracity of either of the two captions.

The challenge of identifying out-of-context samples can be divided into two smaller tasks: assessing the similarity between images and captions and determining the semantic relationship between two captions. For the first task, we propose leveraging scene graph and developing a method for matching these graphs since they have demonstrated their effectiveness in finding the similarity between image-caption pairs [2–4]. As for the second task, we applied natural language inference to capture the semantic relation between two sentences. Natural language inference is more suitable for this task than the semantic textual similarity (STS) score used in the COSMOS baseline, especially in cases where two sentences contradict each other but still exhibit a high STS score. Details of our proposed method are described in section 3.4.

3.2 Context label generation

Labeled datasets for fake news detection are often quite small and focus on limited topics, like the testing set of this challenge. Consequently, supervised learning models tend to overfit and might not work well with unseen data. Although there are sizable news datasets that cover a wide variety of topics like the train and validation sets of the challenge, they typically consist of genuine articles and lack specific labels. Therefore, we propose various manipulation techniques for this task to create labeled data from unlabeled data. We first select two captions corresponding to an image to create a NOOC sample. The OOC samples are created in three different ways:

1. Select two captions that correspond to two different images.
   To compare two images, we extract the visual features using the Resnet-50 model pre-trained by a contrastive learning method named SimCLR [43]. Contrastive learning is suitable for measuring image similarity since it aims to maximize the similarity between similar instances (samples augmented from the same image) and minimize the similarity between dissimilar ones (different images).
   Next, we compare the images by calculating the cosine distance between the extracted two feature vectors. We set a threshold of 0.5, and if the cosine distance value exceeds that threshold, the two images are seen as different.
2. Changing the named entities in the captions
   If two nearly similar captions refer to different events or people, they are definitely "out-of-context". Therefore, we replace the named entities in a caption previously identified by Spacy NER with new named entities that are different from the original ones. We checked carefully to ensure that the new entities differ from those they replaced since a specific name or date can have multiple formats.
3. Adding declarative phrases
   We added some declarative phrases like "This is fake news", "The information is fabricated", ... to the caption to create a new caption that contradicts the original one.

3.3 Scene graph extraction

3.3.1 Visual scene graph

We used the Neural Motifs [22] and PE-Net [31] model to extract visual scene graphs. Neural Motifs have been widely used for text-to-image retrieval and image-to-image retrieval tasks, while PE-Net is a new method that has achieved better results on various benchmarks. We use the two methods separately and then compare the results with each other.

3.3.2 Textual scene graph

To extract textual scene graphs from the captions, we first used the SPICE [33] technique to extract textual scene graphs from the modified captions. SPICE technique can produce quality outcomes with no grammar errors since this is a rule-based method, but there are some instances where it cannot extract scene graphs. Specifically, SPICE cannot extract textual scene graphs from 11.79% of captions from our labeled training sets. If SPICE fails, we use the FACTUAL-T5 [36] model to parse scene graphs from the modified sentences. However, this model can produce a lot of errors like spelling errors or concatenated words. Therefore, we must apply some rules to cope with these errors. If the FACTUAL-T5 model still cannot parse scene graphs from the modified captions, we will use this model to parse scene graphs from the original captions. We prioritize the modified captions over the original captions since visual scene graph parsers cannot recognize specific people or places in the pictures. Therefore, using entity types like "PERSON" and "ORG" to compare the captions with the images is better than using specific names or destinations.

3.4 Model architecture

Our proposed pipeline is comprised of five main components: visual graph encoder, textual graph encoder, graph embedding module, NLI module, and classification module. First, the visual and textual encoder modules encode the scene graphs extracted from the image and the pairs of captions into feature graphs. The feature graphs are then embedded by the Siamese embedding module to get the feature vectors. The two captions are also fed to the NLI module to extract another feature vector. Finally, the classification module combines outputs from the graph embedding and the NLI modules to predict the binary label following two different approaches. The workflow of the architecture is depicted in fig 1.

3.4.1 Visual scene graph encoder module

Label embedding The semantic labels of the objects and relations in the scene graphs are embedded into vectors using two separate embedding layers. Each label of an object or a relation is embedded into a 300-dimensional vector. We initialize the embedding layers using the pre-trained GloVe [44] model.

Image features extraction A pre-trained CNN-based model can be used to extract the visual feature vector from the cropped region of a detected object. Similarly, the visual feature vector can be obtained for an edge by utilizing the union region of the two objects in the scene graph.

Fusion layer We used a fully connected neural network to combine the label features and image features extracted and get a fused feature vector for each node and edge.

Graph Convolutional Network We employ the graph convolutional network (GCN) to update the fused features of nodes and edges in the graph. The GCN model updates each node separately without information from neighboring nodes to avoid noise effects. Conversely, each edge is updated with the information from its two neighboring nodes to learn their relationship as the edge connects these two nodes. Finally, the GCN model outputs the updated features of nodes and edges in the fused features graph. Specifically, for an n-layer GCN model, the updated node features $h^l_{o_i}$ and edge features $h^l_{p_{ij}}$ at the lth layer can be achieved as follows:

$$h^l_{o_i} = MLP_o(h^{l-1}_{o_i}), \quad (1)$$

$$h^l_{p_{ij}} = MLP_p(h^{l-1}_{p_{ij}}), \quad (2)$$

where MLP_o and MLP_p are two different multilayer perceptron models. Initially, $h^0_{o_i} = u_{o_i}$ and $h^0_{p_{ij}} = u_{ij}$, with u_{o_i} and u_{ij} being the fused features of the nodes and edges of the graph that were extracted from the fusion layer.

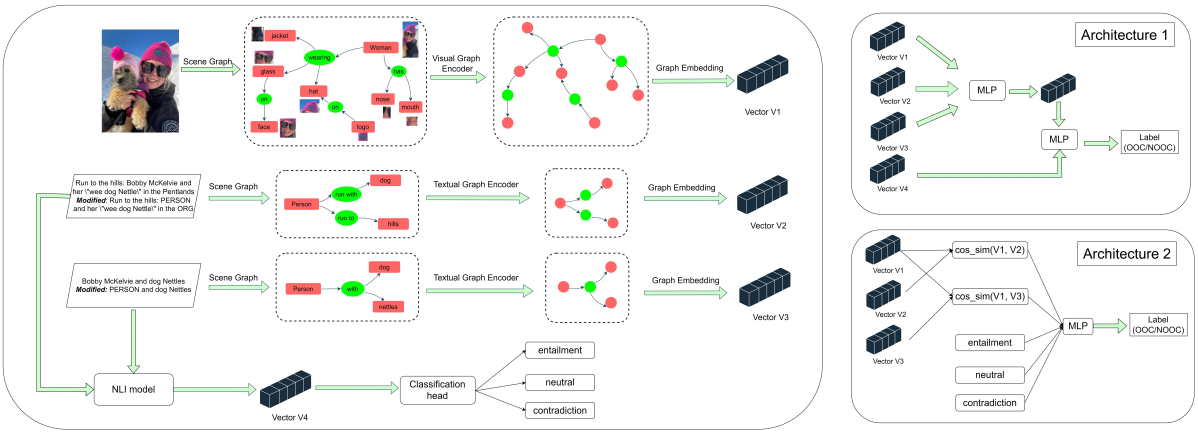


Fig. 1 Our proposed pipeline for the CheapFakes detection problem. First, the scene graphs of an image and two captions are extracted and then encoded to get the feature graphs. The three feature graphs are then embedded into three vectors. In addition, two captions are also put in the NLI model, which will return a vector or three separate values. Next, output from the NLI model and feature vectors are used in two different ways. For the first architecture, the feature vectors from the graph and the NLI model are combined by two multi-layer perceptron (MLP) models to predict the label. The second architecture calculates the cosine similarities between embedded feature vectors and uses them along with NLI scores as input for the MLP model to predict the final results.

3.4.2 Textual scene graph encoder module

Label embedding The label embedding phase in the textual scene graph encoder is similar to the visual modality. During this phase, we replace words that appear only once with the "unknown" word to avoid overfitting.

Graph Network We used bidirectional long-short-term memory (BiLSTM) to learn the features of the words in the triplet and the features of the entire triplet. Next, we adopted a GCN model with a similar structure to the one in the visual module to get the features of nodes and edges in the graph.

3.4.3 Graph embedding layer

After getting updated by the encoder modules, the visual and textual graph features are embedded into vector form using an attention mechanism called multi-scale node attention [45]. This attention structure is shared as a Siamese network with visual and textual feature graphs. Given the set of node features $\{h_1, ..., h_N\}$ and edge features $\{r_1, ..., r_M\}$, the embedded vector g is obtained as follows:

$$g^h = \sum_{n=1}^{N} \sigma(h_n^T ReLU(W_h(\frac{1}{N}\sum_{n=1}^{N} h_m))h_n, \quad (3)$$

$$g^r = \sum_{n=1}^{M} \sigma(r_n^T ReLU(W_r(\frac{1}{M}\sum_{n=1}^{M} r_m))r_n, \quad (4)$$

$$g = [g^h, g^r] \quad (5)$$

where σ and ReLU stands for Sigmoid and rectified linear unit activate functions, W_h and W_r are the parameters in the model.

3.4.4 NLI module

Besides matching the images with the captions, Task 1 of the Cheapfakes detection problem also requires comparing the semantic meaning of the two captions. Therefore, we apply a pre-trained language model for the NLI task to determine the contradiction level of the two captions. Normally, the model will return three values: entailment, neutral, and contradiction. However, the model can return a vector if its classification head is removed. Our model takes the two captions as input and returns a vector or three single values, depending on the architecture. We choose several language models that were pre-trained on

the MultiNLI dataset [46], including BART [47], RoBERTa [48] and DeBERTaV3 [49] models.

3.4.5 Classification module

After extracting feature vectors from the scene graphs and the language model, they can be combined to predict the final results using two different architectures:

(i) The first architecture utilizes two multi-layer perceptron (MLP) models. Specifically, the vectors embedded in the three scene graphs are concatenated and put into the first MLP model, which will return a vector. This first MLP model aims to extract the multimodal feature vector for the image-caption matching task. Next, the second MLP model will take the feature vectors generated by the first MLP model and the NLI language model as input to predict the final result.

(ii) In the second architecture, we determine the two similarity values between the image and the two captions by calculating the cosine similarities between the visual and two textual graph embedding vectors. We also use the entailment, neutral, and contradiction values between the two captions instead of the NLI feature vector like the first architecture. Finally, these five values are fed into an MLP model to predict the classification label.

4 Experiments

4.1 Dataset

We used the dataset for Task 1 from the IEEE ICME 2023 Grand Challenge on Detecting Cheapfakes. The challenge provides the train, validation, and public testing set but hides the private testing set. Each image is associated with one or more captions in the train and validation set but lacks context labels. Only the public testing set is provided with context labels, indicating whether the pair of captions is out-of-context with the image. Each sample in the testing set comprises one image, two captions, and a context label. The number of images in the public testing set is significantly fewer than the train and validation set, and detailed statistics of the challenge dataset are depicted in Table 1 below.

Regarding the training data, we created four different training sets with context labels from the challenge's unlabeled training data. The four sets have 12000 samples, 16000 samples, 80000 samples, and 160000 samples, respectively. We named the datasets D_1, D_2, D_3, and D_4, and detailed information about these training sets is depicted in Table 2 below.

Table 1 Challenge dataset statistics

Split	Images	Captions	Context label
Train	161752	360749	No
Validation	41006	90036	No
Public test	1000	2000	Yes
Private test	N/A	N/A	N/A

Table 2 Statistics of our generated training sets

Type	D_1	D_2	D_3	D_4
NOOC	6000	8000	40000	80000
OOC_1	4000	4000	20000	40000
OOC_2	2000	2000	10000	20000
OOC_3	0	2000	10000	20000
Total	12000	16000	80000	160000

The NOOC represents the number of samples with the NOOC label, while the OOC_1, OOC_2, and OOC_3 represent the three different methods to create the OOC label. The first dataset, D_1, is the only set that does not have out-of-context samples created by adding declarative phrases. The training sets we generated have a similar data format to the public testing set from the challenge.

4.2 Experiment settings

We employ two methods to generate visual scene graphs: Neural Motif and PE-Net. The SPICE technique and the FACTUAL-T5 model are combined to extract textual scene graphs from the captions. For the NLI module, we tested three language models: BART, RoBERTa, and DeBERTaV3. We will experiment with various settings for the visual scene graph extraction method and the NLI module to find the most appropriate one for the problem. We run the experiments on a computer that is equipped with the Intel I7-10700 CPU, 32GB RAM and the RTX3080 GPU.

4.3 Evaluation metrics

Since the problem of identification out-of-context samples can be formulated as a binary classification problem, we will use common metrics for the classification tasks like Accuracy, Precision, Recall, and F1-score to evaluate the results. True Positive (TP) represents a correct identification, False Positive (FP) means an incorrect identification and False Negative (FN) indicates that the target was missed. Precision measures the proportion of TP predictions out of all positive predictions made by the model. It indicates how many of the instances predicted as positive are actually positive. Recall is used to identify all true positive instances. It shows how many of the actual positive instances were correctly identified by the model. F1-score is the harmonic mean of precision and recall, providing a balanced measure between two metrics.

4.4 Experimental results

We trained the models on four training sets and evaluated the results on the public testing set using the F1-score metric. The evaluation results of the model we built with different settings are shown in Table 3.

The method using Pe-NET, RoBERTa model, and the first architecture produced the best results when training on the D_4 set that has 160000 samples. Hence, we chose the results generated by this setting to compare with other methods for the cheapfake detection problem. The models are trained on the train set and evaluated on the public testing set of the challenge dataset.

Besides the COSMOS baseline, we also compare the results with other methods:

- Spotfake [50], which is a multi-modal framework for fake news detection that utilizes a pre-trained language model (like BERT) and VGG-19 to exploit text and image features from an article.
- Self-Query Adaptive-Context Learning [14] that utilizes image search queries to enrich its knowledge about the image and verify the use of captions.
- EANN [51], an end-to-end framework that can extract event-invariant features, thereby enhancing fake news detection on newly arrived events.
- MUICG [52], which focuses on using image captioning to assess the correlation between images and captions.
- Knowledge graph [16], an approach that utilizes knowledge graph to learn more information from external knowledge.

We use four common metrics: F1-score, precision, recall, and accuracy. Evaluation results are displayed in Table 4.

4.5 Discussion

This section discusses the results of the proposed method. Firstly, in Table 3, we can see the results of Approach 1 consistently outperforms Approach 2 across all models. Specifically, the F1 score in Approach 1 reaches up to 85.69%, whereas most results in Approach 2 range only from 71% to 76%. Moreover, the D_4 dataset proves to be superior to the other datasets in Approach 1, achieving results above 81% across all models. The BART and DeBERTa model shows that using Neural Motifs to extract visual scene graphs has higher results compared to PE-Net in Approach 1. Conversely, for the RoBERTa model, PE-Net shows better performance. Specifically, with the D_4 dataset, the F1-score increases by 2.3% compared to Neural Motifs, and there are increases of 3.61% and 3.9% with the D_2 and D_3, respectively. This means the setting with the PE-Net and RoBERTa model shows a significant enhancement, indicating that this pairing is particularly effective. The data shows that using PE-Net for visual scene graph extraction along with the RoBERTa model consistently leads to higher accuracy and better overall performance. This combination outperforms other model and dataset pairings.

From the evaluation results, one can see that the first architecture, which leverages all the extracted feature vectors, produces better results than the second one, which relies solely on single values when we scale up the number of training samples. This suggests that calculating only the cosine similarities between the embedded vectors and the entailment, neutral, and contradiction values might not be sufficient to assess whether the triplet is out of context or not. Using the feature vector can provide richer information, therefore achieving higher results. Changes in the visual scene graph generation method and the

Table 3 Evaluation results using F1-score metric with different settings.

Approach	Data	BART		RoBERTa		DeBERTa	
		PENET	Neural Motifs	PENET	Neural Motifs	PENET	Neural Motifs
1	D_1	0.7246	0.7327	0.7487	0.7522	0.7344	0.7886
	D_2	0.7961	0.8110	0.8060	0.7668	0.8004	0.8186
	D_3	0.8016	0.8127	0.8088	0.7727	0.8156	0.8227
	D_4	0.8176	0.8216	**0.8569**	0.8330	0.8231	0.8320
2	D_1	0.7242	0.7281	0.7266	0.7230	0.7185	0.7165
	D_2	0.7406	0.7383	0.7235	0.7222	0.7184	0.7159
	D_3	0.7533	0.7699	0.7334	0.7465	0.7578	0.7540
	D_4	0.7322	0.7342	0.7194	0.7192	0.7231	0.7305

Table 4 Evaluation results of different methods for the cheapfakes detection in Task 1

Method	F1-score	Precision	Recall	Accuracy
Spotfake [50]	0.5350	0.5252	0.5306	0.5279
Self-Query [14]	0.6106	0.5907	0.6320	0.5970
EANN [51]	0.6073	0.6025	0.6122	0.6300
MUICG [52]	0.7940	0.7885	0.7995	0.7930
COSMOS [1]	0.8100	**0.8520**	0.7720	0.8190
Knowledge graph [16]	0.8405	0.8407	0.8405	0.8405
CSGM (ours)	**0.8569**	0.8460	**0.8680**	**0.8550**

NLI models have little impact on the prediction outcomes.

The results show that our model outperforms previous methods in Table 4, including the COSMOS baseline. Specifically, our method achieves an F1-score of 85.69% and an accuracy score of 85.5%, compared to the COSMOS baseline, which has an F1-core of 81% and an accuracy of 81.9%. This represents an increase of 4.69% in the F1-score and 3.6% in accuracy, highlighting our model's superior performance. Furthermore, when considering recall, our model maintains its superiority. With a recall of 86.8%, it significantly surpasses the COSMOS baseline, which achieved a recall of 77.2%. This substantial difference underscores the superior ability of our model to identify and cover true positive cases within the dataset accurately.

5 Conclusion and future work

In our study, we used scene graph matching to detect cheap fakes and performed experiments with different settings to confirm the effectiveness of our approach. We also developed augmentation techniques to create labeled data from unlabeled data. Our proposed method does not need to be fine tuned to the public testing set or explicit data; instead, it utilizes the training set provided by the challenge. The experimental results have shown that the proposed model achieved competitive results and had higher performance than the COSMOS baseline. In the future, we will explore more methods to extract scene graphs from images and captions and find more effective ways to generate labeled data, like applying large language models (LLM). Also, we plan to enhance the modeling part based on new features extracted.

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